# Number Theory 2024 Final Solutions

1. Where $n = 1105$, we find that $n-1 = 1104 = 2^4 \cdot 69$. We compute, the following modulo 1105. Fill in the chart with $Y$ or $N$ signifying which values of $a$ are Euler witnesses and which are Miller Rabin witnesses (witnessing the fact that $n$ is composite)?

| $a$ | $a^{69}$ | $a^{2\cdot 69}$ | $a^{2^2\cdot 69}$ | $a^{2^3\cdot 69}$ | $a^{2^4\cdot 69}$ | Euler witness? | MR witness? |
|---|---|---|---|---|---|---|---|
| 15 | 70 | 480 | 560 | 885 | 885 | | |
| 47 | 47 | $-1$ | 1 | 1 | 1 | | |
| 66 | 716 | 1048 | 781 | 1 | 1 | | |
| 81 | 846 | 781 | 1 | 1 | 1 | | |

**Solution**

| $a$ | $a^{69}$ | $a^{2\cdot 69}$ | $a^{2^2\cdot 69}$ | $a^{2^3\cdot 69}$ | $a^{2^4\cdot 69}$ | Euler witness? | MR witness? |
|---|---|---|---|---|---|---|---|
| 15 | 70 | 480 | 560 | 885 | 885 | $Y$ | $Y$ |
| 47 | 47 | $-1$ | 1 | 1 | 1 | $N$ | $N$ |
| 66 | 716 | 1048 | 781 | 1 | 1 | $N$ | $Y$ |
| 81 | 846 | 781 | 1 | 1 | 1 | $N$ | $Y$ |

2. Show that no Carmichael number is the product of exactly two primes. (If you can't do this, define Carmichael numbers and state Korselt's Criterium for partial marks.)

**Solution**

(A composite number $n$ is Carmichael if $a^n \equiv_n a$ for all residues $a$ mod $p$. Korselt: composite $n$ is Carmichael if and only if it is a product of distinct odd primes $p$ such that $p-1|n-1$.)

If $n = pq$ were Carmichael then by Koselt's Criterium we have that $p-1|n-1$ and may assume that $q < p$. But then $p-1$ divides $n-1 = pq-1$ and so divides $(pq-1)-(p-1)q = pq-1-pq+q = q-1$. But that $p-1$ divides $q-1$ contradicts $q < p$.

3. Show, modulo some odd prime $p$, that the product of two $QR$ is a $QR$ and that the product of a $QR$ and an $NR$ is an $NR$. (Or for partial marks, define $QR$ and $NR$.)

**Solution**

(A number $N \in \{1, \dots p-1\}$ is a $QR$ mod $p$ if there is some $n$ such that $N \equiv_p n^2$. It is a $NR$ if there is no such $n$.)

If $A$ and $B$ are $QR$ then there are $a$ and $b$ such that $A \equiv_p a^2$ and $B \equiv_p b^2$ so $AB \equiv a^2b^2 = (ab)^2$ showing that $AB$ is also a $QR$; this gives the first part. If $A$ is a $QR$ and $B$ an $NR$ then $AB$ is an $NR$. Indeed, if it were a $QR$, then where $A \equiv_p a^2$, we have

$$c^2 \equiv_p AB \equiv_p a^2 B.$$

But as $a$ is invertible, being non-zero mod $p$, we get that

$$B \equiv_p \left(\frac{c}{a}\right)^2.$$

This contradicts the fact that $B$ is an $NR$.

4. **By computing Legendre Symbols,** determine if 41 and 97 are QR modulo the prime $p = 101$.

**Solution**

41 isn't and 97 is:

$$\begin{aligned}\left(\frac{41}{101}\right) &= \left(\frac{101}{41}\right) = \left(\frac{19}{41}\right) = \left(\frac{41}{19}\right) = \left(\frac{3}{19}\right) \\ &= -\left(\frac{19}{3}\right) = -\left(\frac{1}{3}\right) = -1\end{aligned}$$

and

$$\left(\frac{97}{101}\right) = \left(\frac{-4}{101}\right) = \left(\frac{-1}{101}\right)\left(\frac{4}{101}\right) = \left(\frac{-1}{101}\right) = 1.$$

5. Using that the result is true when $b$ is an odd prime, show for any odd number $b$ that $\left(\frac{2}{b}\right) = 1$ if $b \equiv_8 1$ or 7, and that $\left(\frac{2}{b}\right) = -1$ if $b \equiv_8 3$ or 5.

**Solution**

For an odd prime $b$ we have that modulo 8, $b$ is in $\{\pm 1, 4 \pm 1\}$ we show that $\left(\frac{2}{b}\right) = 1$ iff $b \equiv_8 \pm 1$. Indeed, let

$$b = p_1 \cdots p_s \cdot q_1 \cdots q_t$$

be the prime decomposition of $b$, where $p_i \equiv_8 \pm 1$ for each $i$ and $q_j \equiv_8 4 \pm 1$ for each $j$. By definition of the Jacobi symbol we have

$$\left(\frac{2}{b}\right) = \prod_{i=1}^{s} \left(\frac{2}{p_i}\right) \prod_{j=1}^{t} \left(\frac{2}{q_j}\right) = (1)^s(-1)^t.$$

On the other hand, computing modulo 8, we have that $p_1 p_2 \ldots p_s \equiv_8 \pm 1$ and observing that

- $\pm 1 \cdot \pm 1 \equiv_8 \pm 1$
- $(4 \pm 1) \cdot (4 \pm 1) \equiv_8 \pm 1$ and
- $\pm 1 \cdot (4 \pm 1) \equiv_8 (4 \pm 1)$

we get that $b \equiv_8 \pm(q_1 q_2 \cdots q_t) \equiv_8 \pm 1$ if and only if $t$ is even. This is exactly when $\left(\frac{2}{b}\right) = 1$, as needed.

6. Observe that $50^2 + 1^2 = 41 \cdot 61$. **Using the method of descent**, write a smaller multiple of 61 as a sum of two squares.

**Solution**

Where $A = 50$ and $B = 1$ we reduce them symmetrically modulo 41 to get $a = 9$ and $b = 1$, giving us

$$9^2 + 1^2 = a^2 + b^2 = 2 \cdot 41.$$

Using the equation $(a^2 + b^2)(A^2 + B^2) = (aA + bB)^2 + (bA - aB)^2$ we get

$$(2 \cdot 41)(61 \cdot 41) = (9^2 + 1^2)(50^2 + 1^2) = (450 + 1)^2 + (50 - 9)^2.$$

Dividing through by $41^2$ we get

$$11^2 + 1^2 = 2 \cdot 61.$$

7. Which of the following numbers can be the hypontenuse $c$ in a primitive pythagorean triple $(a, b, c)$:

   (a) $754 = 2 \cdot 13 \cdot 29$,

   (b) $1131 = 3 \cdot 13 \cdot 29$,

   (c) $1885 = 5 \cdot 13 \cdot 29$?

**Solution**

We know that $c$ has to be a product of (odd) primes congruent to 1 mod 4. So only $\boxed{1885}$ will do.

8. (a) Define *primitive root modulo* $p$.
   (b) How many primitive roots are there modulo 47?
   (c) Find a primitive root modulo 13.
   (d) Show that if $a$ is a primitive root modulo $p$ and $\gcd(d, p-1) = 1$, then $a^d$ is also a primitive root modulo $p$.

**Solution**

(a) A *primitive root modulo* $p$ is an element $a$ with order $e_p(a) = p - 1$; ie, such that the minimum power $e$ such that $a^e \equiv_p 1$ is $e = p - 1$.
(b) $\phi(46) = \phi(2)\phi(23) = 22$.
(c) If $a$ isn't primitive mod 13 then $a^6 \equiv_{13} 1$. Checking $2^6 \equiv_{13} -1$, we get that 2 is primitive.
(d) We need to show that the order $e = e_p(a^d)$ is $p - 1$. By Fermat's Little Theorem and the Order Divisibility Theorem, $e \mid p - 1$. On the other hand $1 \equiv_p (a^d)^e = a^{e \cdot d}$, so by the order divisiblity theorem, as $e_p(a) = p - 1$, we have $(p - 1) \mid e \cdot d$. As $\gcd(p - 1, d) = 1$, this implies $(p - 1) \mid e$; so $e = p - 1$, as needed.

(So anything in $\{2, 2^5, 2^7, 2^{11}\} = \{2, 6, 11, 7\}$ is an answer to (c).)

9. (a) What is a Costas array of size $n$?
   (b) Show that $(g^1, \ldots, g^{p-1})$ modulo $p$ is a Costas array of size $p - 1$ if $g$ is a primitive root modulo $p$.

**Solution**

i) It is a permutation $(a_1, a_2, \ldots, a_n)$ of the integers $[n]$ such that there are no $d \in [n - 1]$ and $i < j \in [n]$ such that $a_{i+d} - a_i = a_{j+d} - a_j$.
ii) If it is not a Costas array, then there are $d$ and $i < j$ such that $g^{i+d} - g^i = g^{j+d} - g^j$, and so

$$g^i(g^d - 1) = g^{i+d} - g^i = g^{j+d} - g^j = g^j(g^d - 1).$$

But as $d < p - 1$, and $g$ is a primitive element, $g^d - 1 \not\equiv_p 0$, and so we can cancel $g^d - 1$ and get $g^i = g^j$. As $g$ is a primitive this means that $i = j$, which is a contradiction.

10. Using the index table below, solve the congruences
   (a) $14x \equiv_{31} 15$
   (b) $3x^{17} \equiv_{31} 41$.

   Modulo $p = 31$ for generator $g = 3$.

| $a$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $3^a$ | 3 | 9 | 27 | 19 | 26 | 16 | 17 | 20 | 29 | 25 | 13 | 8 | 24 | 10 | 30 |
| $a$ | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| $3^a$ | 28 | 22 | 4 | 12 | 5 | 15 | 14 | 11 | 2 | 6 | 18 | 23 | 7 | 21 | 1 |

| $a$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $I(a)$ | 30 | 24 | 1 | 18 | 20 | 25 | 28 | 12 | 2 | 14 | 23 | 19 | 11 | 22 | 21 |
| $a$ | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| $I(a)$ | 6 | 7 | 26 | 4 | 8 | 29 | 17 | 27 | 13 | 10 | 5 | 3 | 16 | 9 | 15 |

**Solution**

We use the generator $g = 3$.
Part i) the index equation is

$$I(x) \equiv_{30} I(15) - I(14) \equiv_{30} 21 - 22 \equiv_{30} 29.$$

So $\boxed{x = 21}$.

Part ii) Reducing mod 31 the equation is $3x^{17} \equiv_{31} 10$. The index equation is

$$17I(x) \equiv_{30} I(10) - I(3)14 - 1 = 13.$$

Using the EEA we get the inverse of 17 modulo 30 to be $-7$:

| *Step* | $a_i$ | $b_i(q_i)$ | $r_i$ | $x(30)$ | $y(17)$ |
|---|---|---|---|---|---|
| 1 | 30 | $17(-1)$ | 13 | 1 | $-1$ |
| 2 | 17 | $13(-1)$ | 4 | $-1$ | $1 + 1 = 2$ |
| 3 | 13 | $4(-3)$ | 1 | $1 + 3 = 4$ | $-1 - 3(2) = -7$ |

Thus $I(x) \equiv_{30} 13(-7) = -91 \equiv_{30} -1 \equiv_{30} 29$ and so $\boxed{x = 21}$.